Bias and Representation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Teach bounded comparisons and explicit representation evidence.

Supply non-identifying fixtures, preserve group sizes and assess a fresh rate comparison.

Keep one habit: Ask what the labels mean, which task failed and what evidence would be needed for a causal or fairness conclusion.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Explain the **mechanism** and terminology of **representation** and bias evaluation

Two 45-minute sessions. Bias evaluation tests defined outcomes across relevant conditions. Group averages can expose differences but cannot alone establish their cause or the correct remedy.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

By the end, I can explain bias and **representation** using a traced example and a stated limitation.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Lesson and workbook; paper and pencil
- Calculator and optional local Python 3 for arithmetic; no live API or private data
- Prepare the fresh retry separately and withhold its solution until after collection

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Copyright and Provenance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of copyright and permitted sources

Reference example, not the only acceptable wording: Teacher explanation: Permission to access a work and permission to redistribute or modify it are separate.

Use a classroom licence ledger instead of assuming all online material is reusable. For this fictional policy, licence L permits classroom display with attribution but forbids public redistribution.

Licence M permits adaptation and redistribution with attribution. Unknown permission means seek approval or choose a permitted alternative; this exercise is not a legal ruling.

Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of red team exercises

Reference example, not the only acceptable wording: Teacher explanation: Red-team exercises probe explicit failure boundaries in an authorised isolated fixture environment. Their purpose is **evidence**-based improvement.

Use fictional records and allowed test routes. Define a target contract violation, record the exact case, observe action/data traces and preserve a reproducible report. Do not touch external systems.

Without opening the old lesson, explain one key idea from Prototype a Grounded Assistant. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of capstone grounded assistant

Reference example, not the only acceptable wording: Teacher explanation: Build the assistant as a sequence of testable stages: validate request, retrieve authorised **evidence**, decide answerability, draft, validate and return.

Avoid merging all errors into one model score. A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims.

The classroom implementation performs no booking or external action.

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Copyright and Provenance

Red Team the Product

Prototype a Grounded Assistant

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

An illustration set always shows scientists in the same role and appearance. Define a review that examines **representation** without inferring personal identities from images.

Concept to explain

Outputs should be tested for omission, stereotyping, and uneven quality.

Term	Meaning
representation	A way to encode information so a person or system can work with it.
stereotype	An oversimplified expectation about members of a group that ignores individual variation.
subgroup quality	Performance measured for a defined subset using stated criteria and denominators.
audit	A systematic examination of records and behaviour against stated checks.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach representation, task quality and denominators together

> Do not infer demographic identity or causation from arbitrary case labels.

Read the audit contract

Audit question	What to preserve
Task and outcome	What counts as supported, useful or complete for this question?
Groups or conditions	Why are these labels relevant; what do they actually mean?
Denominators	How many cases in each group and how many pass?
Error types	Which exact claims, omissions or fallbacks failed?
Case comparability	Difficulty, source coverage, prompt wording and annotation rules
Bounded conclusion	Observed difference or missing coverage; no invented causal story
Proposed revision	One controlled change and a fresh matched retest plan, not an invented improvement

Bias evaluation tests defined outcomes across relevant conditions. Group averages can expose differences but cannot alone establish their cause or the correct remedy.

Define the task, relevant groups, labels and denominators before comparing. Use fictional, non-identifying records. Evaluate quality and fallback outcomes separately; preserve sample sizes and unresolved explanations.

Trace separate A/B support rates

A grounded assistant supports 9 of 10 group-A cases and 6 of 10 group-B cases.

Rates 90% and **60%**; gap 30 percentage points. Investigate **case mix**, retrieval coverage and annotation rules before attributing cause.

Read the Birch/Pine record

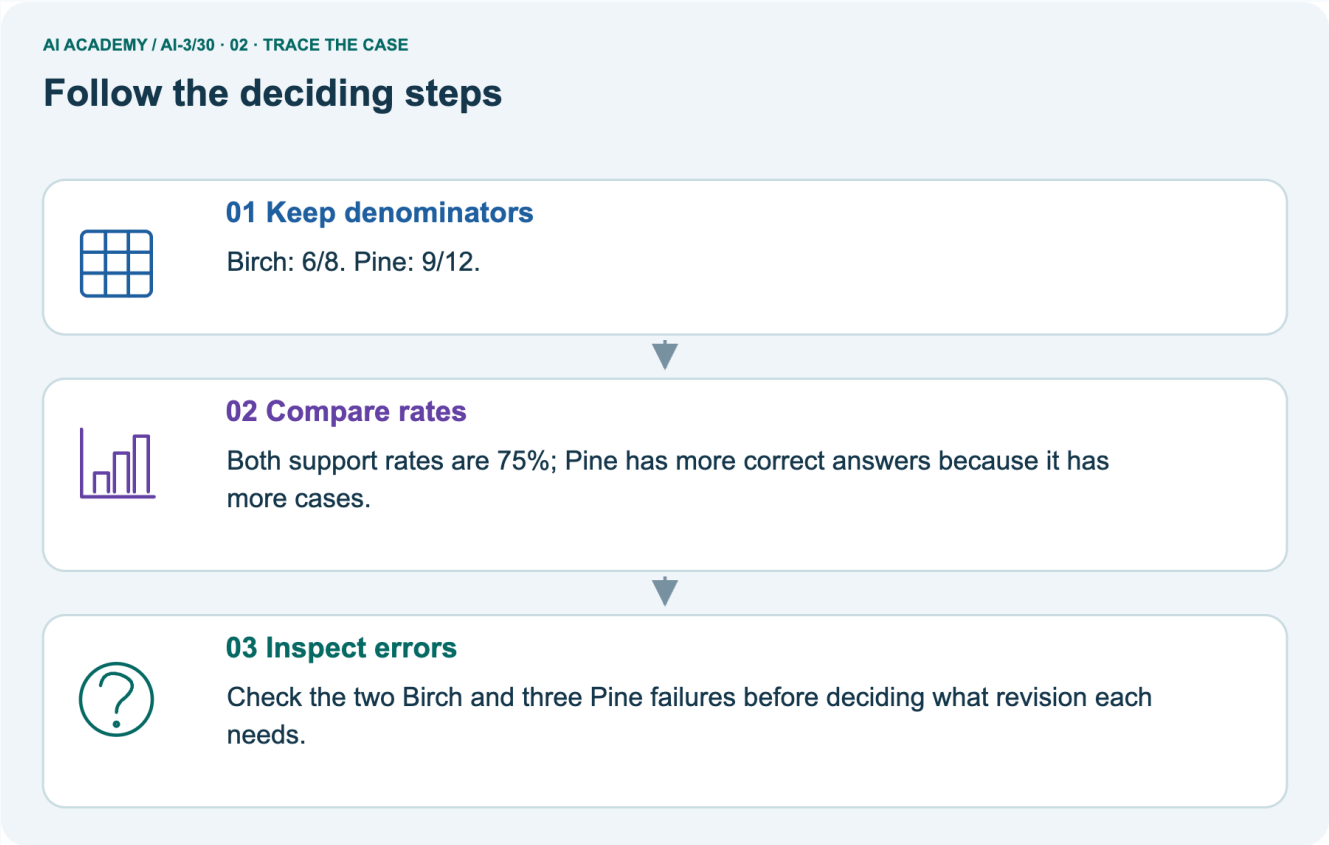
Compare study-answer support rates when groups have different numbers of questions.

Group Birch has 6 supported answers in 8 cases; group Pine has 9 supported answers in 12 cases. Both groups use the same support rubric.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Birch and Pine are arbitrary fictional case-group labels, not demographic descriptions. Birch 6/8 and Pine 9/12 both equal 75%; failures are 2 and 3. Changed Pine 6/12 is 50%, 25 percentage points below Birch. A percentage-point gap subtracts rates: 75%-50%=25 points; it is not the same as a relative percent change. No source coverage, difficulty matching or causal mechanism is supplied.

Trace equal rates and different failure counts



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep denominators	Birch: 6/8. Pine: 9/12.
2. Compare rates	Both support rates are 75%; Pine has more correct answers because it has more cases.
3. Inspect errors	Check the two Birch and three Pine failures before deciding what revision each needs.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Check the two Birch and three Pine failures before deciding what revision each needs. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict the changed Pine rate

Change: Pine instead has 6 supported answers out of 12.

Prediction: Pine is 50%, which is 25 percentage points below Birch.

Reason: Equal correct counts do not imply equal rates when denominators differ.

State comparability and generalisation limits

These groups are fictional; small samples and unmatched question difficulty limit conclusions about fairness.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a **representation**; it does not document execution of a real AI model.

▼ Additional discussion · Representation requires declared criteria

Scientist images repeatedly show only one demographic group.

Evaluate and improve **representation**.

Review relevant **representation** criteria across **output**s, revise inclusive requirements and retest. Appearance alone cannot establish absence of all bias.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Generate comparable characters while changing one identity descriptor.

2. Count omissions, **stereotype**d roles, and quality differences.

3. A balanced count does not alone prove respectful depiction.

4. Repair prompt, data, filters, or use limits, then repeat the fixed **audit**.

Recalculate, retrace, or restate the reasoning in your own **representation**:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Exact explanation to model

Explain the added worked example

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Subgroup rate	A measured outcome within a stated group
Percentage-point gap	The difference between two percentage rates
Case mix	The distribution of task conditions in a group

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

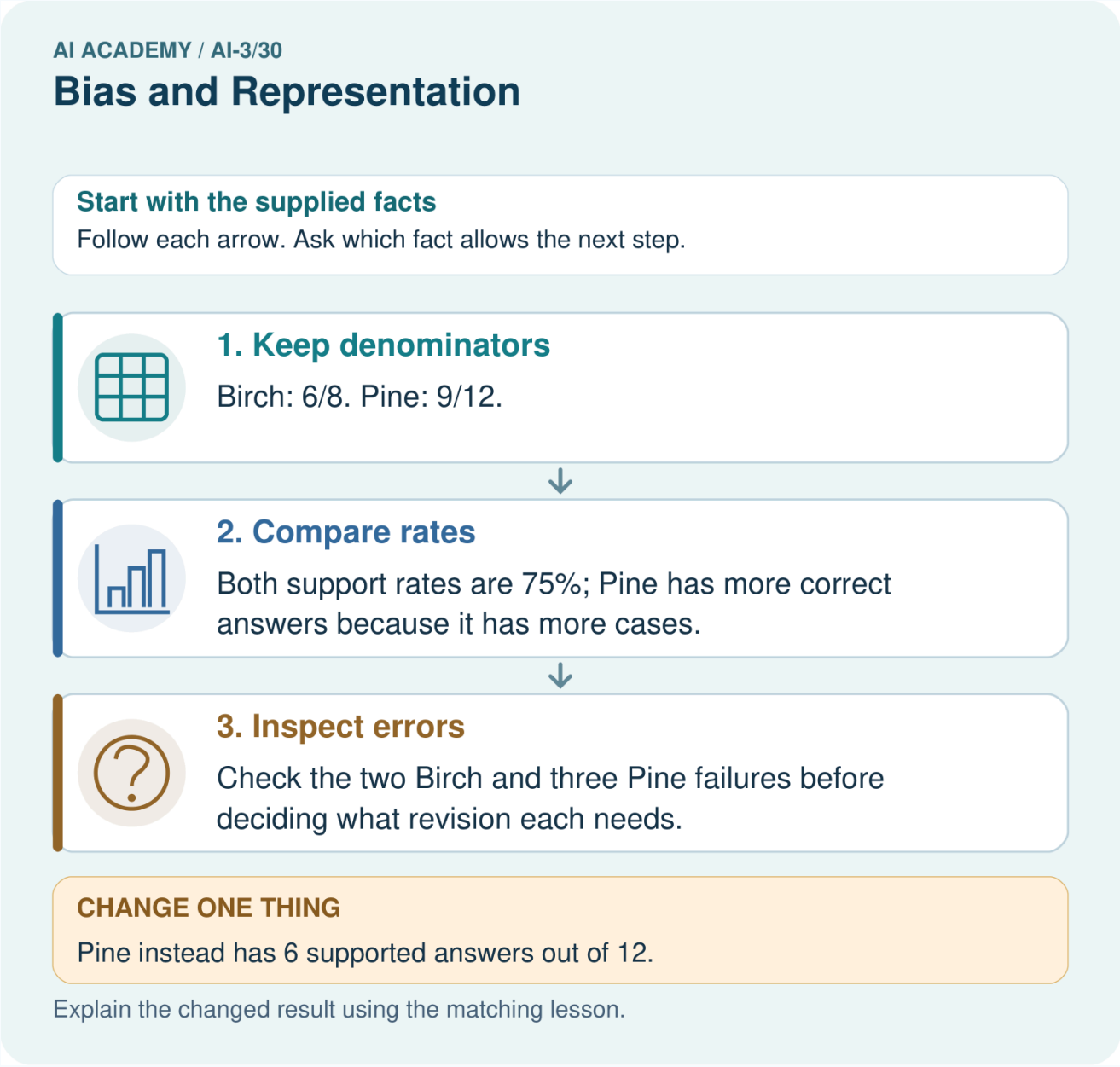
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or **audit**ory learning styles.

A pictorial worked case: Bias and Representation

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the explanation, then remove help

> Collect each learner’s reasoning before showing feedback.

Main completion prompt

Completion task: A grounded assistant supports 9 of 10 group-A cases and 6 of 10 group-B cases. Show the intermediate steps and give the conclusion with its scope.

Expected reasoning and fading help

Teacher check: Teacher explanation: Rates 90% and **60%**; gap 30 percentage points. Investigate **case mix**, retrieval coverage and annotation rules before attributing cause.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Representation** and bias evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Group comparisons require task relevance and appropriate data governance; differences alone do not identify a causal **mechanism**.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Explain why comparing six correct answers in each group would mislead. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Birch has 6/8=75%; changed Pine has 6/12=50%. Counts must be paired with the number of opportunities. Changed-case check: Pine is 50%, which is 25 percentage points below Birch.

Equal correct counts do not imply equal rates when denominators differ.

Changed-case reasoning: Pine is 50%, which is 25 percentage points below Birch. Equal correct counts do not imply equal rates when denominators differ.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: Scientist images repeatedly show only one demographic group.

Write the question to explain: Evaluate and improve **representation**.

Reveal after **prediction**s: Review relevant **representation** criteria across **output**s, revise inclusive requirements and retest. Appearance alone cannot establish absence of all bias.

Use the earlier changed question as optional later practice. Reserve Compare the fresh unequal-size C/D groups independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: Rates 90% and **60%**; gap 30 percentage points. Investigate **case mix**, retrieval coverage and annotation rules before attributing cause.

Qualification: Group comparisons require task relevance and appropriate data governance; differences alone do not identify a causal **mechanism**.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.

Stage	Teacher action
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Birch/Pine base 75% each with 2 and 3 failures; revised Pine 50%, 25 percentage points below Birch. A/B W2 is 90%/60%, 30 points. Paired character cards each pass 2/3 task-fit checks, with different failures; no model tendency or causal gender effect is measured. Independent C 8/10=80%, D 9/15=60%, gap 20 points; higher raw successes in D do not imply higher rate. Preserve small-sample and case-match limits.

Run and record

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Use the complete fictional Amal/Rowan paired-prompt fixture and six constructed outputs. Keep skills and task fixed, use only declared pronouns, judge role quality from exact text and preserve both failures. Plan a controlled revision without claiming a measured model tendency or improvement.

4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

The worked result establishes every future case: Group comparisons require task relevance and appropriate data governance; differences alone do not identify a causal **mechanism**.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Compare the fresh unequal-size C/D groups independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: Group C has 8 supported answers out of 10; D has 9 out of 15. Compute rates and difference; explain why raw correct totals mislead.

Teacher explanation: C=**80%**, D=**60%**; gap 20 percentage points. D has more supported answers in count but lower rate. No causal explanation follows from the counts.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Bias evaluation tests defined outcomes across relevant conditions. Group averages can expose differences but cannot alone establish their cause or the correct remedy.

Define the task, relevant groups, labels and denominators before comparing. Use fictional, non-identifying records. Evaluate quality and fallback outcomes separately; preserve sample sizes and unresolved explanations.

W2 Trace the worked case

A grounded assistant supports 9 of 10 group-A cases and 6 of 10 group-B cases. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Rates 90% and **60%**; gap 30 percentage points. Investigate **case mix**, retrieval coverage and annotation rules before attributing cause.

W3 Audit the limit

A report omits this qualification: Group comparisons require task relevance and appropriate data governance; differences alone do not identify a causal **mechanism**.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Group comparisons require task relevance and appropriate data governance; differences alone do not identify a causal **mechanism**.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Group comparisons require task relevance and appropriate data governance; differences alone do not identify a causal **mechanism**.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Review relevant **representation** criteria across **output**s, revise inclusive requirements and retest. Appearance alone cannot establish absence of all bias.

Additional transfer practice question

An assistant is tested on six fluent requests but no requests with spelling errors. What gap does this leave, and how should the evaluation expand?

Reasoned practice answer

It cannot support claims about performance on spelling-error requests. Add representative examples and compare outcomes, without treating spelling as **evidence** of ability.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Strong results for one language style prove equal performance for all learners.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Review varied roles, activities and stated **representation** goals across **output**s, avoiding assumptions about hidden identities. Revise requirements and retest.

Visual variety alone does not prove absence of all bias; discuss who may still be missing and why.

Explanations for the additional workbook examples

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Group C has 8 supported answers out of 10; D has 9 out of 15. Compute rates and difference; explain why raw correct totals mislead.

Teacher explanation: C=**80%**, D=**60%**; gap 20 percentage points. D has more supported answers in count but lower rate. No causal explanation follows from the counts.

W5 Design a revealing follow-up

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Audit answer quality across reading levels, languages, names, and accessibility needs.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support while preserving the reasoning goal

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: C=**80%**, D=**60%**; gap 20 percentage points. D has more supported answers in count but lower rate. No causal explanation follows from the counts. Require intermediate reasoning and the stated scope.

Fresh retry: Group E=12/15, F=7/10. Expected reasoning: **80%** versus **70%**, gap 10 percentage points; small constructed samples and possible case differences limit conclusions.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Plan the delayed check

Suggested return points in this level: ai-3/31, ai-3/33, ai-3/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 31: Safety Boundaries. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 31: Safety Boundaries. Carry forward task-specific criteria, relevant conditions and honest uncertainty when defining what a helper must decline, clarify or route to a person.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Verified Study Buddy

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Explain the **mechanism** and terminology of **representation** and bias evaluation

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

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> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Practice and assessment reminders

Before class, solve the supported practice: A grounded assistant supports 9 of 10 group-A cases and 6 of 10 group-B cases. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Rates 90% and **60%**; gap 30 percentage points. Investigate **case mix**, retrieval coverage and annotation rules before attributing cause.

Check every learner before discussion: Use the worked case for **Representation** and bias evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Group comparisons require task relevance and appropriate data governance; differences alone do not identify a causal **mechanism**.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Pine instead has 6 supported answers out of 12. Require a **prediction** and a reason before comparing. Expected result: Pine is 50%, which is 25 percentage points below Birch.

Reason: Equal correct counts do not imply equal rates when denominators differ.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

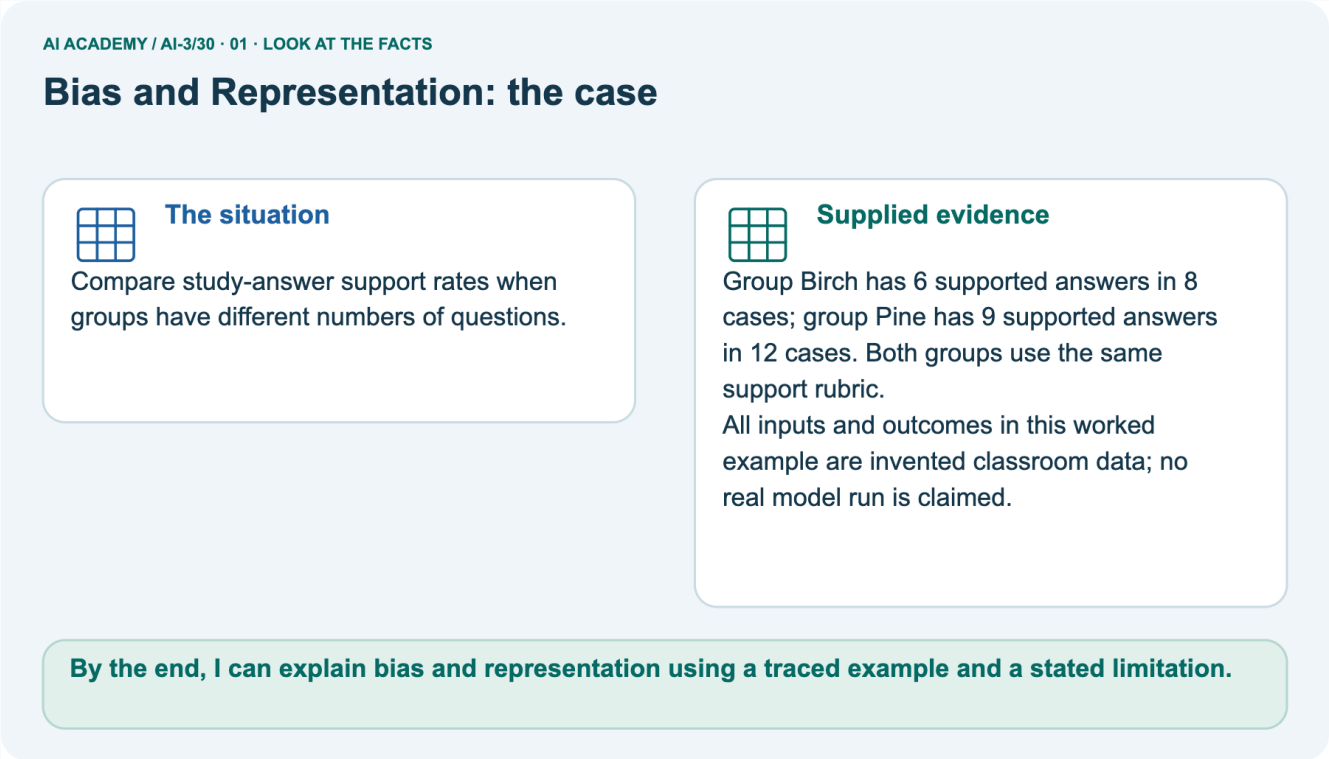
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Bias and Representation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

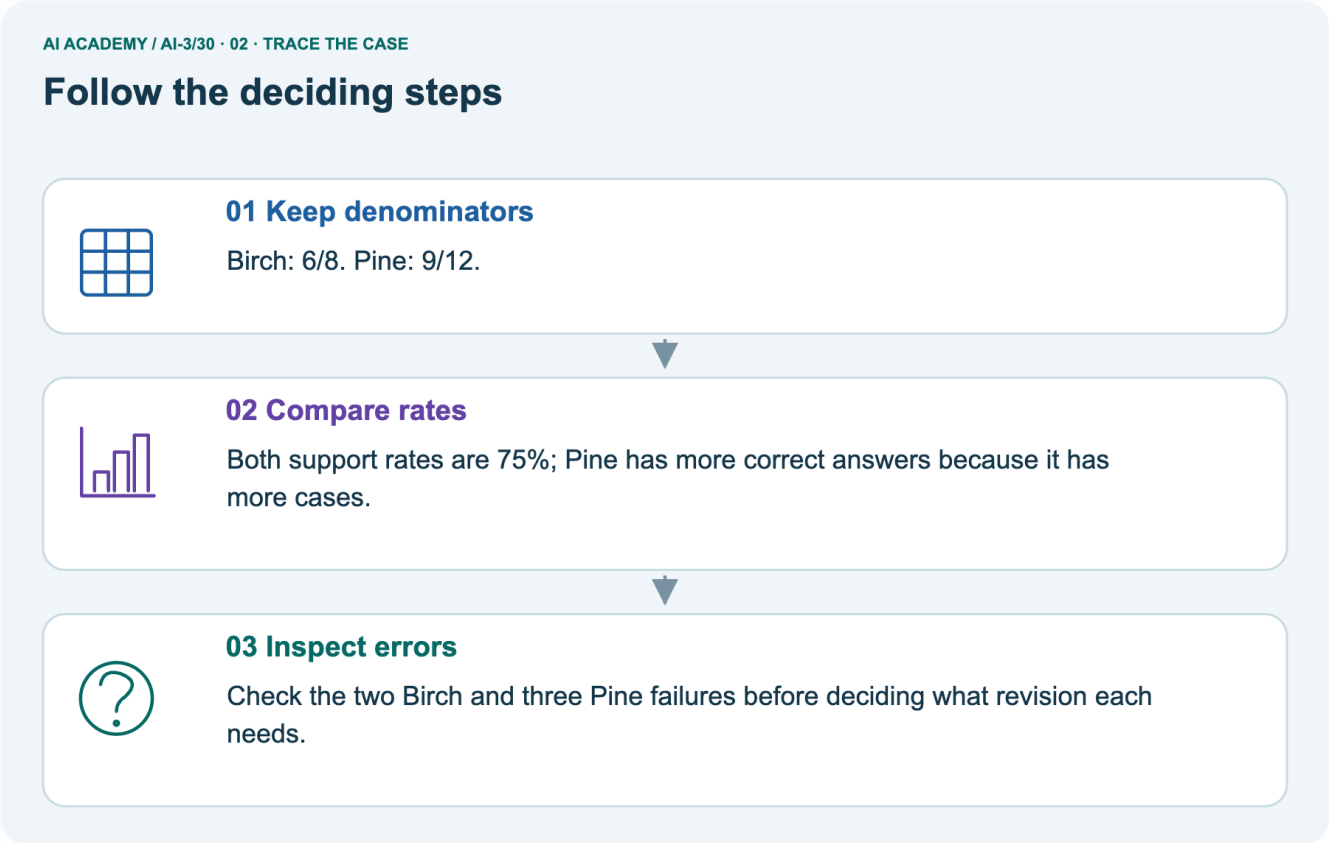
Compare study-answer support rates when groups have different numbers of questions. Supplied facts: Group Birch has 6 supported answers in 8 cases; group Pine has 9 supported answers in 12 cases.

Both groups use the same support rubric. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

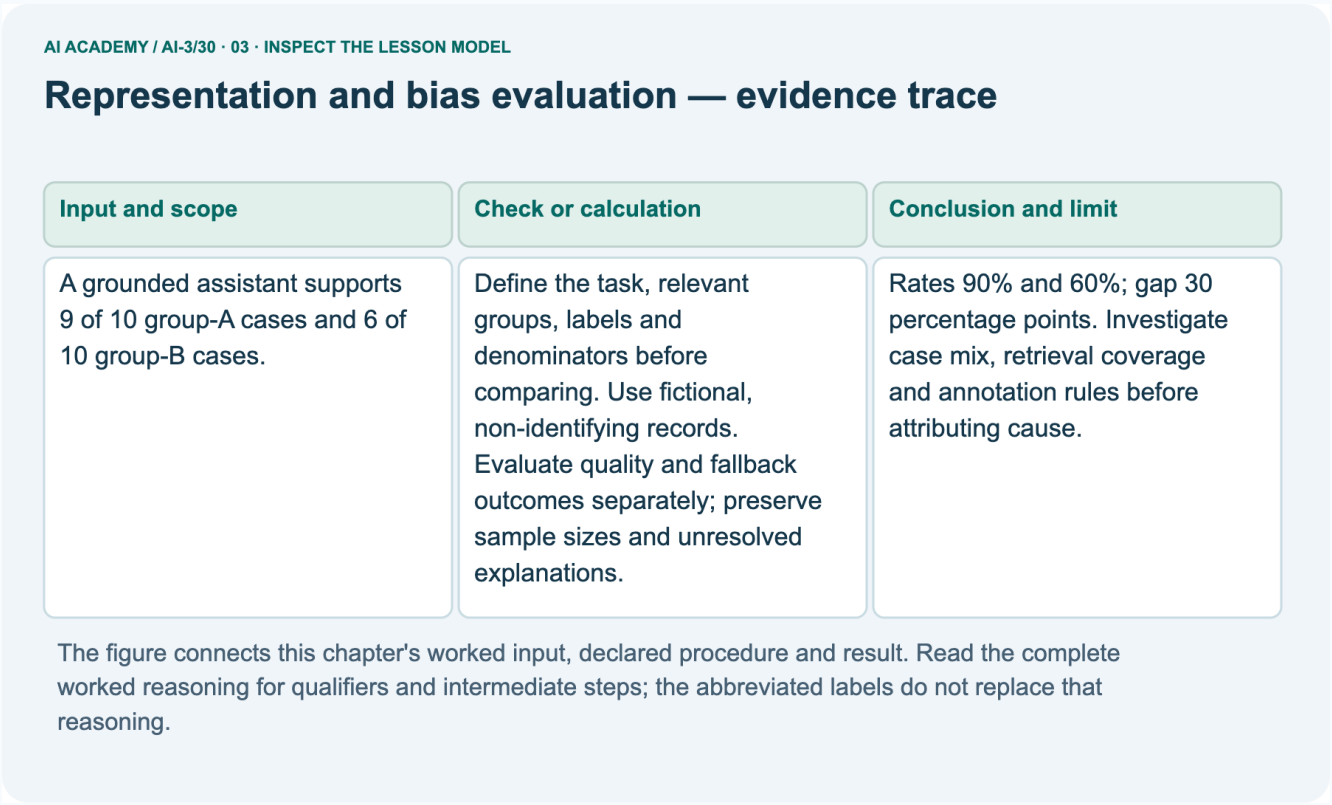
Keep denominators: Birch: 6/8.

Pine: 9/12. → Compare rates: Both support rates are 75%; Pine has more correct answers because it has more cases. → Inspect errors: Check the two Birch and three Pine failures before deciding what revision each needs.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Representation and bias evaluation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Representation and bias evaluation — **evidence** trace. A grounded assistant supports 9 of 10 group-A cases and 6 of 10 group-B cases. | Define the task, relevant groups, labels and denominators before comparing.

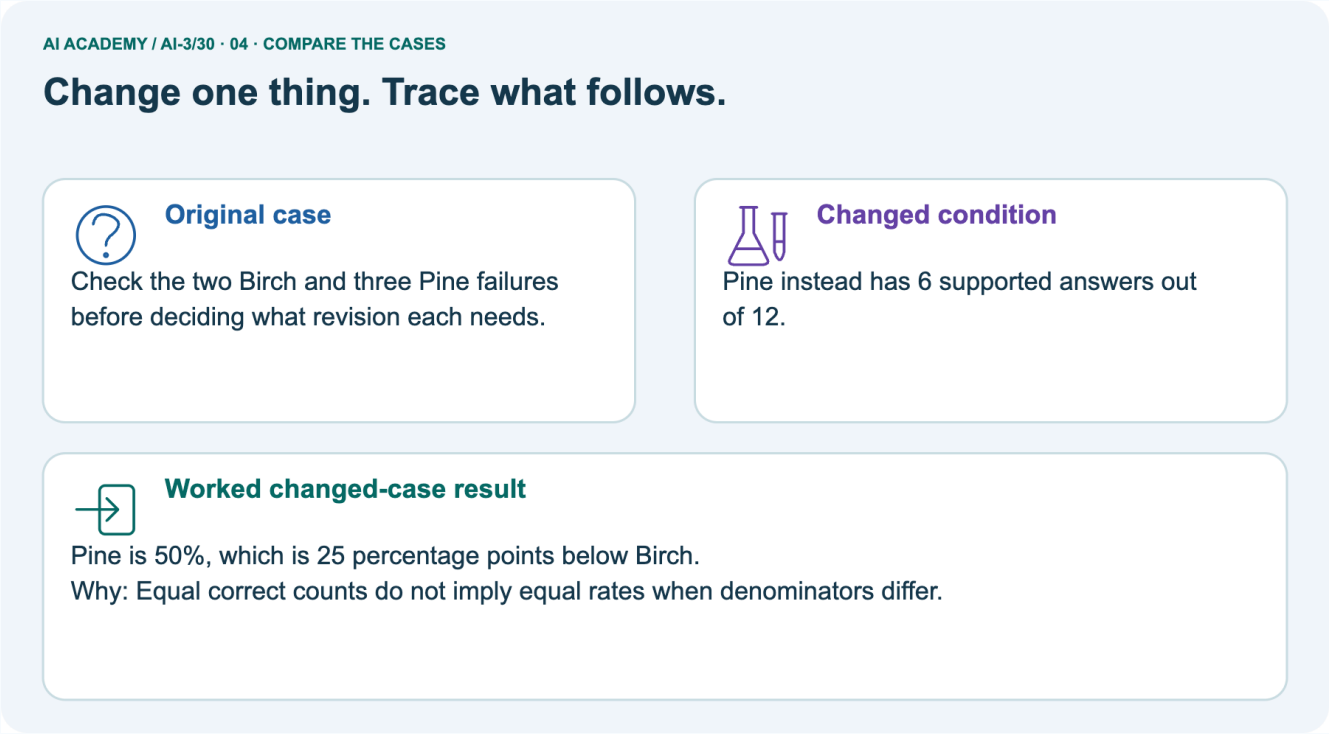
Use fictional, non-identifying records. Evaluate quality and fallback outcomes separately; preserve sample sizes and unresolved explanations. | Rates 90% and **60%**; gap 30 percentage points.

Investigate **case mix**, retrieval coverage and annotation rules before attributing cause.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Check the two Birch and three Pine failures before deciding what revision each needs. Changed condition: Pine instead has 6 supported answers out of 12.

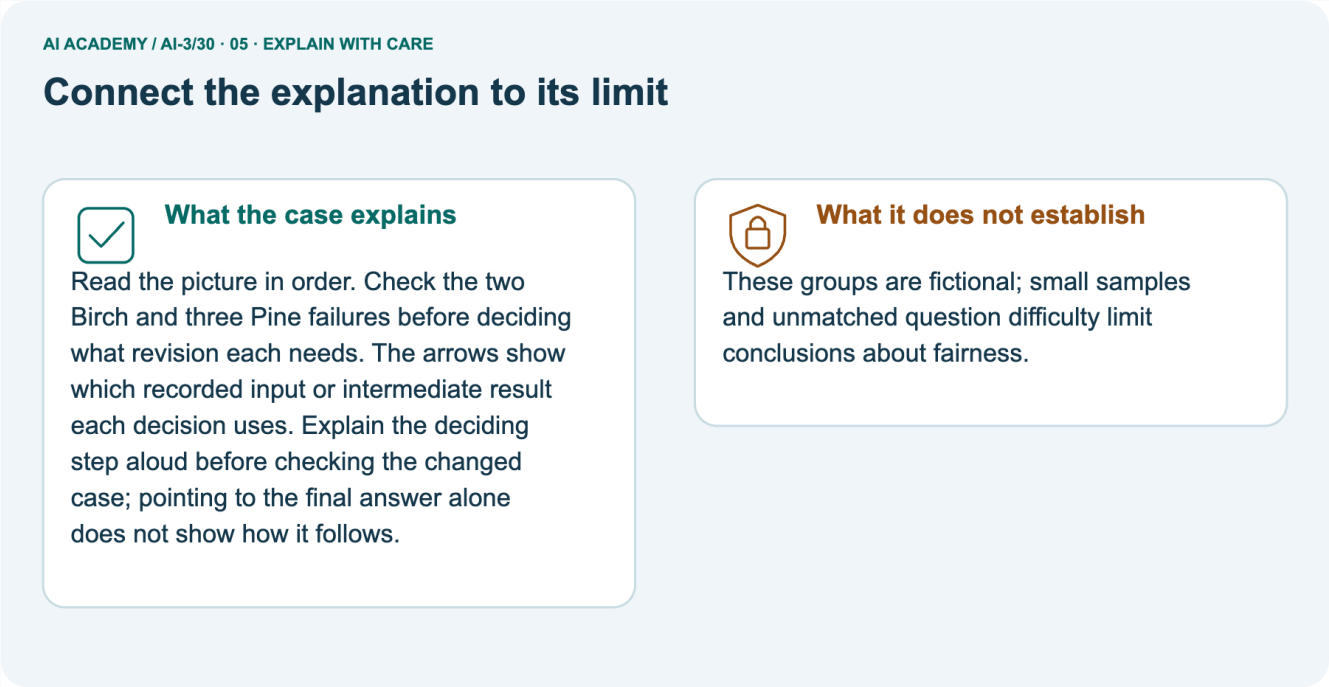
Expected result: Pine is 50%, which is 25 percentage points below Birch. Reason: Equal correct counts do not imply equal rates when denominators differ.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Equal correct counts do not imply equal rates when denominators differ.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Check the two Birch and three Pine failures before deciding what revision each needs. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

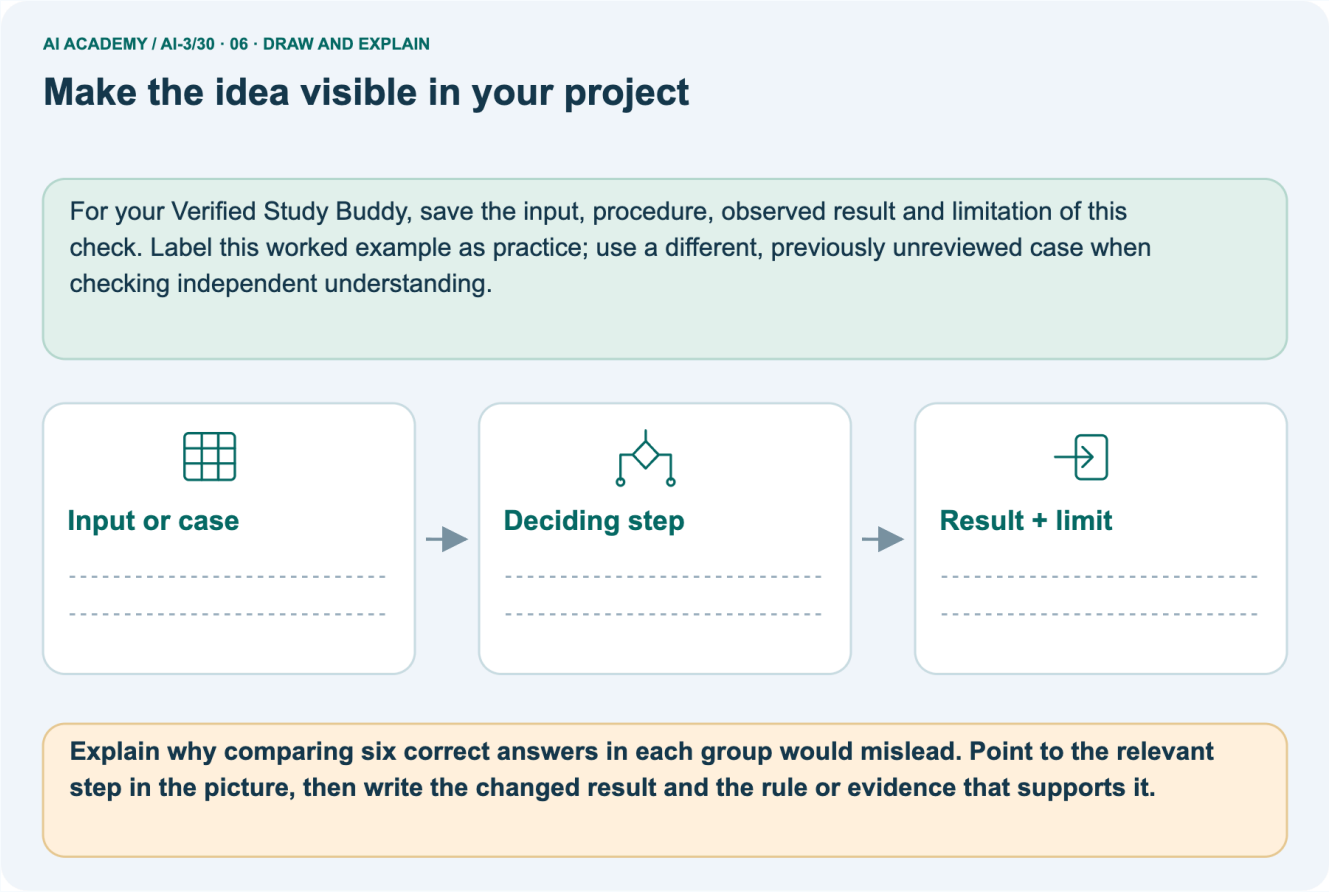
Limit: These groups are fictional; small samples and unmatched question difficulty limit conclusions about fairness.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These groups are fictional; small samples and unmatched question difficulty limit conclusions about fairness.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Explain why comparing six correct answers in each group would mislead. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Birch has 6/8=75%; changed Pine has 6/12=50%. Counts must be paired with the number of opportunities. Changed-case check: Pine is 50%, which is 25 percentage points below Birch.

Equal correct counts do not imply equal rates when denominators differ.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Bias and Representation: the case

02. Follow the deciding steps

03. Representation and bias evaluation — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.